CHIRANJEEVI VANTAKU

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PROFESSIONAL SUMMARY

Forward Deployed AI Engineer with 3+ years of experience building LLM applications, RAG pipelines and multi-agent systems. Architected a **healthcare agentic platform** at Medbridge with six specialized agents orchestrated through LangGraph, spanning clinical documentation, care management, and revenue workflows. Implemented policy-grounded RAG that increased ***clinician acceptance of recommendations to over 85%***. Applied generative AI to consumer research and reporting at Beats by Dre, ***reducing reporting turnaround time by 60%***. Combines hands-on AI engineering with **stakeholder collaboration** and end-to-end solution delivery.

PROFESSIONAL EXPERIENCE

Medbridge Inc. Virginia

Forward Deployed AI Engineer **May 2024 - Present**

- Architected a multi-agent healthcare automation platform with 6 specialized agents orchestrated via LangGraph, supporting pre-rounding, post-rounding, chronic care management (CCM) analysis, clinical documentation, benefits verification, prior authorization and billing reducing administrative workload by 40+ hours/week, with workflows validated by clinicians and billing teams.
- Engineered an AI-powered ambient clinical scribe integrating streaming speech recognition with LLM-driven clinical language processing to transform patient-provider conversations into structured, codified SOAP notes, with automated diagnosis and procedure mapping to ICD-10 and CPT codes.
- Delivered a payment reconciliation pipeline that compares electronic remittance advice (ERA) against contracted fee schedules, identifies underpayments, and assembles supporting evidence for billing-team review and dispute resolution.
- Built an HCC risk-adjustment gap detector that scans diagnosis histories for chronic conditions documented in prior years but missing from the current year (RAF-impacting recapture), flagging 420+ patients with unclosed HCC codes and surfacing ~$180K in value-based reimbursement at risk.
- Developed an AI-powered care-gap analysis agent that evaluates patient demographics, ICD-10 diagnosis history, payer information, and prior CPT/HCPCS claims against evidence-based rules across 40+ preventive and chronic-care gap types, routing actionable recommendations to medical assistants and providers with last-completed dates and supporting clinical codes.
- Implemented a RAG layer over CMS and payer policy documents (NCD/LCD coverage rules, preventive-service schedules) so coding recommendations cite the governing policy passage - grounding AI outputs in authoritative sources and raising clinician acceptance of suggestions to 85%+.
- Developed a real-time AI voice agent for patient-access calls using WebRTC, handling appointment scheduling, refill requests, and benefits questions with Tebra integration, verified identity checks, emergency escalation, and clinician handoff, reducing front-desk call volume by 47% across multiple clinics.

Beats by Dre New York

Artificial Intelligence Engineer **Dec 2024 - May 2025**

- Engineered RAG pipelines over consumer research, product documentation, and market reports, integrating semantic chunking, hybrid retrieval, and source attribution to deliver grounded research insights.
- Developed multi-agent workflows coordinating retrieval, analysis, and validation, with shared context, task delegation, and failure recovery to automate research synthesis.
- Built MCP servers and tool-calling integrations connecting AI assistants to document repositories, SQL databases, and business APIs, enabling evidence-backed responses across structured and unstructured data.
- Implemented domain-specific LLM fine-tuning and evaluation pipelines, benchmarking adapted models against baseline performance for factual accuracy, instruction adherence, and structured output quality.
- Owned end-to-end AI application delivery, translating stakeholder requirements into deployed workflows with automated evaluations, execution tracing, and latency and token-cost monitoring.
- Automated AI-powered research synthesis and executive reporting using Claude and Gamma, transforming consumer insights and performance findings into stakeholder presentations and product strategy recommendations, reducing reporting turnaround time by 60%.

Columbia University New York
Machine Learning Researcher **Sep 2024 - Nov 2024**

- Investigated MM-Diffusion architecture to generate synchronized high-fidelity audio-video content by leveraging novel denoising and Random-Shift-based Multi-Modal Attention mechanisms, enhancing cross-modal alignment and coherence while reducing computational redundancy.
- Published research findings showing 28% better FVD scores and 31% improved FAD metrics on AIST++ dataset, outperforming 3 state-of-the-art baseline models.
- Optimized multi-modal attention mechanism that decreased training time by 50% while maintaining 94% generation quality, potentially saving $15K in GPU costs.

PUBLICATIONS

High-Dimensional Audio Genre Classification Using MFCC-Driven Multi-Layer Perceptrons. **[Paper]**

- Published in IRJET, Volume 12, Issue 05 | May 2025. Developed a deep learning-based music genre classifier using MFCCs and MLPs, achieving 64% validation accuracy on the GTZAN dataset through dimensionality reduction and regularization techniques.

RADAR: Radiology Appropriateness Decision & Advisory Resource **[Paper]**

- Built RADAR, a retrieval-augmented clinical decision-support system grounding Claude Opus 4.8 in ACR Appropriateness Criteria at variant-level granularity - combining BM25 retrieval, an evidence-gated abstention mechanism, and a deterministic citation validator that achieved 100% topic accuracy, 100% citation fidelity, and 7/7 off-topic abstention across ED consult scenarios, with sub-millisecond guardrail overhead; open-sourced code and evaluation harness

SKILLS

Languages & APIs: Python, SQL, C, FastAPI, API Integration, API Testing, Model Context Protocol (MCP).

Generative AI & Agents: LLM Applications, RAG Pipelines, Semantic Chunking, Hybrid Retrieval, Embeddings, Vector Search, Prompt Engineering, LLM Fine-Tuning, Multi-Agent Orchestration, Tool Calling, Shared Context Management, LangChain, LangGraph, LlamaIndex, CrewAI, Hugging Face, PyTorch, vLLM.

Evaluation & Observability: LLM Evaluation, Factual Accuracy and Groundedness Assessment, Structured Output Validation, Regression Testing, Agent Execution Tracing, Latency and Token-Cost Monitoring, MLflow.

Data & Retrieval: MySQL, SQL Server, BigQuery, Amazon Redshift, Vector Databases, pandas, NumPy, PySpark, ETL/ELT Pipelines, Apache Airflow, Document Ingestion.

Cloud & Deployment: Docker, Kubernetes, GitHub, MLOps, Google Cloud Platform (Vertex AI, Compute Engine), AWS (EC2, S3, Lambda, Glue).

Client Delivery: Stakeholder Collaboration, Requirements Gathering, Solution Architecture, Workflow Integration, Technical Presentations, End-to-End AI Application Delivery.

EDUCATION

Stevens Institute of Technology, Hoboken, NJ **Sept 2024 - May 2026**
Master of Science in Computer Science **GPA:** 3.9/4.0